

ITO5201 Cheatsheet

1 Sample Statistics

- quartiles, percentiles, etc.: given n data points, rank them in increasing value to get $x_1, \dots, x_n$
 - median, if n is odd, given by $x_{(n+1)/2}$, if n is even given by $\frac{1}{2}(x_{n/2} + x_{n/2+1})$
 - quartiles, Q_1 or Q_3 is given by $Q_k = x_p + \frac{q}{4}(x_{p+1} - x_p)$ where $p = \text{floor}((k(n+1))/4)$ and $q = (k(n+1)) \bmod 4$
 - percentiles, $P_k = x_p + \frac{q}{100}(x_{p+1} - x_p)$ where $p = \text{floor}((k(n+1))/100)$ and $q = (k(n+1)) \bmod 100$
- measures of spread for n data points, $\mathbf{x} = (x_1, \dots, x_n)$
 - Sample variance, $\text{var}(\mathbf{x}) = s_x^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2$
 - Sample standard deviation, s_x
 - Range = $\max_{i=1}^n x_i - \min_{i=1}^n x_i$
 - (inter-quartile range) IQR = $Q_3 - Q_1$

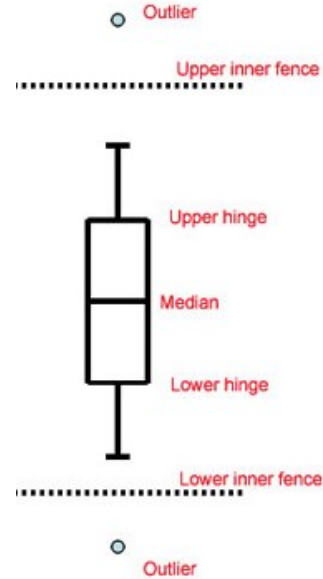
- Sample covariance

$$q_{xy} = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})$$

- Sample correlation coefficient

$$r_{xy} = \frac{q_{xy}}{s_x s_y} = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^n (x_i - \bar{x})^2 \sum_{i=1}^n (y_i - \bar{y})^2}}$$

- boxplots
 - represents numerical data through quartiles
 - the lower hinge is Q_1 , upper hinge is Q_3
 - a lower whisker is drawn at minimum data value greater than the lower inner fence is $Q_1 - 1.5 \times \text{IQR}$ (which itself is usually not draw)
 - upper whisker is drawn at maximum data value less than the upper inner fence is $Q_3 + 1.5 \times \text{IQR}$ (which itself is usually not draw)
 - outliers are highlighted outside these two fences



2 Probability

- probability axioms of Kolmogorov:
 1. for any event A , $0 \leq p(A) \leq 1$
 2. $p(\Omega) = 1$, where Ω is the universal set, the set of everything
 3. for mutually exclusive events $A_1, \dots, A_n$ $p(A_1 \cup A_2 \dots \cup A_n) = \sum_{i=1}^n p(A_i)$
- other probability identities for the domain $\mathcal{X} \times \mathcal{Y}$ where A, B are any events:
 - complement rule, $p(\bar{A}) = 1 - p(A)$
 - product rule, $p(B \cap A) = p(B|A)p(A)$
 - sum rule, $p(A) = \sum_{x \in \mathcal{X}} p(x \cap A)$
 - conditional, $p(B|A) = \frac{p(B \cap A)}{p(A)}$ when $p(A) > 0$
 - Bayes theorem, $p(x|A) = \frac{p(A|x)p(x)}{\sum_{x \in \mathcal{X}} p(A|x)p(x)}$

- for continuous random variables a **probability density function** (PDF) $p(x)$ on domain $\mathcal{X}$ satisfies

$$p(x) \geq 0 \text{ for all } x \in \mathcal{X}$$

and

$$\int_{\mathcal{X}} p(x) dx = 1$$

- for continuous random variables, the probability $X \in A$, where $A \subset \mathcal{X}$ is

$$p(X \in A) = \int_A p(x) dx.$$

- for $\mathcal{X}$ a single dimension, then define the **cumulative density function** (CDF), $P(x)$, in terms of the the PDF $p(x)$ as

$$P(x) = \int_{y < x} p(y) dy$$

and the **quantile function** $Q(x)$ as

$$Q(x) = P^{-1}(x)$$

this is well defined when $p(x) > 0$.

- Let the random variable pair (X, Y) be from domain $\mathcal{X} \times \mathcal{Y}$. We say X and Y are **independent** if any of the following three (equivalent) conditions hold for all $x \in \mathcal{X}$ and $y \in \mathcal{Y}$

$$\begin{aligned} (I) \quad p(X=x|Y=y) &= p(X=x) && \text{when } p(Y=y) > 0 \\ (II) \quad p(Y=y|X=x) &= p(Y=y) && \text{when } p(X=x) > 0 \\ (III) \quad p(Y=y \cap X=x) &= p(X=x)p(Y=y) \end{aligned}$$

3 Expected Values

- if X has domain $\mathcal{X}$, **expectation** and **variance** of $f(X)$:

$$\mathbb{E}[f(X)] = \sum_{x \in \mathcal{X}} p(x) f(x)$$

$$\mathbb{V}[f(X)] = \mathbb{E}[(f(X) - \mathbb{E}[f(X)])^2] = \mathbb{E}[f(X)^2] - \mathbb{E}[f(X)]^2$$

with integral replacing sum for continuous RVs

- define **entropy** (to base 2 by default)

$$H(X) = \mathbb{E} [\log_2 1/p(X)]$$

- some properties of entropy where X has discrete domain $\mathcal{X}$:
 - if $\mathcal{X}$ of finite dimension K , then $0 \leq H(X) \leq K$
 - if $H(X) = 0$ then $p(X=x) = 1$ for some $x \in \mathcal{X}$
 - entropy can be justified as the lower bound in bits of a binary prefix code to encode a realisation of X
- a **binary prefix code** for symbols assigns a binary string to every symbol such that no code is the prefix of another
 - given a distribution on K symbols denoted $1, 2, \dots, K$, with probability $p(k) = p_k$, there is a prefix code for the symbols with expected code length $\leq 1 + H(\mathbf{p})$ using code lengths $l_k = \lceil \log_2 1/p_k \rceil$
- some useful rules for RVs X, Y and constant c
 - $\mathbb{E} [f(X) + g(Y)] = \mathbb{E} [f(X)] + \mathbb{E} [g(Y)]$
 - $\mathbb{E} [cf(X)] = c\mathbb{E} [f(X)]$
 - $\mathbb{V} [cf(X)] = c^2\mathbb{V} [f(X)]$
 - $H(X, Y) = H(X) + H(Y|X) = H(Y) + H(X|Y)$
- if X, Y are independent RVs
 - $\mathbb{E} [f(X)g(Y)] = \mathbb{E} [f(X)] \mathbb{E} [g(Y)]$
 - $\mathbb{V} [f(X) + g(Y)] = \mathbb{V} [f(X)] + \mathbb{V} [g(Y)]$
 - $H(X, Y) = H(X) + H(Y)$
- **Chebyshev's inequality**: if X is a RV with mean μ and variance σ^2 , then for any $k > 0$

$$p \left(\frac{|X - \mu|}{\sigma} \geq k \right) \leq \frac{1}{k^2}$$

- **Chebyshev's inequality for samples:** for a sample $S = \{x_1, \dots, x_N\}$ of variable X with mean $\bar{x}$ and sample standard deviation s_x , then for any $k > 0$

$$\left| \left\{ x_i : \frac{|x_i - \bar{x}|}{s_x} \geq k \right\} \right| \leq \frac{N}{k^2}$$

that is, the number of data points at least $k s_x$ from the mean is no more than $\frac{N}{k^2}$.

- **Weak law of large numbers:** let $X_1, \dots, X_n$ be RVs with $\mathbb{E}[X_i] = \mu$; then for any $\varepsilon > 0$

$$p\left(\left|\frac{X_1 + \dots + X_n}{n} - \mu\right| > \varepsilon\right) \rightarrow 0 \text{ as } n \rightarrow \infty.$$

4 Distributions

- for the **Gaussian or normal distribution**, denoted $N(\mu, \sigma^2)$

$$p(x | \mu, \sigma^2) = \left(\frac{1}{2\pi\sigma^2}\right)^{\frac{1}{2}} \exp\left(-\frac{(x - \mu)^2}{2\sigma^2}\right)$$

and has the properties

- $\mathbb{E}[x] = \mu$ and $\mathbb{V}[x] = \sigma^2$
- the mode and the median are the same as the mean
- if the curve for $p(x | 0, 1)$ is shifted to the right by μ and scaled by $1/\sigma$, one gets the curve for $p(x | \mu, \sigma^2)$
- the **Bernoulli distribution** models discrete, binary RVs, i.e., $\mathcal{X} = \{0, 1\}$, denoted $\text{Be}(\theta)$,

$$p(X = 1 | \theta) = \theta, \theta \in [0, 1]$$

so that the parametric probability distribution is

$$p(x | \theta) = \theta^x (1 - \theta)^{(1-x)}$$

and has properties

- $\mathbb{E}[x] = \theta$ and $\mathbb{V}[x] = \theta(1 - \theta)$

- the **binomial distribution** describes the probability of getting x successful outcomes in n Bernoulli trials with probability of success θ , denoted $\text{bin}(\theta, n)$, and $x \in \{0, 1, \dots, n\}$,

$$p(x | n, \theta) = \binom{n}{x} \theta^x (1 - \theta)^{(n-x)}$$

and has properties

$$- \mathbb{E}[x] = n\theta \text{ and } \mathbb{V}[x] = n\theta(1 - \theta)$$

- the **Poisson distribution** with rate parameter λ is the number of events x occurring, for $\mathcal{X} = \{0\} \cup \mathcal{N}$, denoted $\text{Pois}(\lambda)$,

$$p(x | \lambda) = \frac{\lambda^x \exp(-\lambda)}{x!}$$

and has properties

- $\mathbb{E}[x] = \lambda$ and $\mathbb{V}[x] = \lambda$
- if $X \sim \text{Pois}(\lambda_X)$ and $Y \sim \text{Pois}(\lambda_Y)$ then $(X + Y) \sim \text{Pois}(\lambda_X + \lambda_Y)$
- $\text{bin}(\theta, n) \approx \text{Pois}(n\theta)$ for $n \gg 1$ and $n\theta$ small

- **Central Limit Theorem (CLT)**: have distribution with mean μ and variance σ^2 , and sample n identical RVs $X_1, \dots, X_n$ from it; then the sample mean $\frac{1}{n} \sum_{i=1}^n X_i$ is approximately distributed as $N(\mu, \frac{1}{n}\sigma^2)$ for large n
- examples of the CLT
 - it is exact in the case of the Gaussian
 - for the binomial, $\text{bin}(\theta, n) \approx N(n\theta, n\theta(1 - \theta))$ for $n \gg 1$ and θ not near 0 or 1
 - for the Poisson, $\text{Pois}(\lambda) \approx N(\lambda, \lambda)$ for $\lambda \gg 1$

5 Estimation

- have a sample $\mathbf{x}$; let $\hat{\theta}(\mathbf{x})$ be a point estimate for model parameter θ ; then $\hat{\theta}(\mathbf{x})$ is **unbiased** if $\mathbb{E}_{\mathbf{x}}[\hat{\theta}(\mathbf{x})] = \theta$, where the expectation is taken over samples $\mathbf{x}$

- the **bias** of the estimator is

$$b_{\theta}(\hat{\theta}) = \mathbb{E}_{\mathbf{x}} [\hat{\theta}(\mathbf{x})] - \theta$$

- the **variance** of the estimator is

$$\mathbb{V}_{\theta} [\hat{\theta}] = \mathbb{E}_{\mathbf{x}} \left[\left(\hat{\theta}(\mathbf{x}) - \mathbb{E}_{\mathbf{x}} [\hat{\theta}(\mathbf{x})] \right)^2 \right]$$

- the **mean square error** (MSE) of the estimator is

$$\text{MSE}_{\theta} [\hat{\theta}] = \mathbb{E}_{\mathbf{x}} \left[\left(\hat{\theta}(\mathbf{x}) - \theta \right)^2 \right] = b_{\theta}(\hat{\theta})^2 + \mathbb{V}_{\theta} [\hat{\theta}]$$

- for sample $\mathbf{x}$ of size n distributed as $N(\mu, \sigma)$ the **sum of squared errors** (SSE) of mean estimate μ is given by

$$\text{SSE}(\mu) = \sum_{i=1}^n (x_i - \mu)^2$$

and the point estimate $\hat{\mu}$ minimising SSE is the mean

$$\hat{\mu} = \frac{1}{n} \sum_{i=1}^n x_i$$

- the method of maximum likelihood says we should use the model that assigns the greatest probability to the data we have observed; formally, the **maximum likelihood estimator** (MLE) is found by solving

$$\hat{\Theta} = \arg \max_{\Theta} \{p(\mathbf{x} | \Theta)\}$$

where $p(\mathbf{x} | \Theta)$ is called the **likelihood function**

- use $L(\mathbf{x} | \Theta)$ to denote the **negative log-likelihood**, $\log 1/p(\mathbf{x} | \Theta)$
- for sample $\mathbf{x}$ of size n distributed as $N(\mu, \sigma)$

$$L(\mathbf{x} | \mu, \sigma^2) = \frac{n}{2} \log(2\pi\sigma^2) + \frac{1}{2\sigma^2} \text{SSE}(\mu)$$

from this we get

- $\hat{\mu}_{ML}$ is the mean, same as when using the SSE
- the MLE for the variance is

$$\hat{\sigma}_{ML}^2 = \frac{1}{n} \sum_{i=1}^n (x_i - \hat{\mu})^2$$

this is however biased, an unbiased estimate is

$$\hat{\sigma}_u^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \hat{\mu})^2$$

- the MLE estimates for λ of the Poisson and θ of the Bernoulli is also the mean
- the MLE estimates for θ of the binomial, $\text{bin}(\theta, m)$, using sample $\mathbf{x}$ of size n is

$$\hat{\theta}_{ML} = \frac{1}{nm} \sum_{i=1}^n x_i$$

- let $\mathbf{x}$ be a sample of size n from a Gaussian population with mean μ and variance σ^2 , and let m be the mean and s^2 be the sample variance:
 - m is Gaussian with mean, variance $\left(\mu, \frac{\sigma^2}{n}\right)$
 - $\sqrt{n}(m - \mu)/s$ is Student's t with $n - 1$ degrees of freedom
 - $(n - 1)s^2/\sigma^2$ is chi-squared with $n - 1$ degrees of freedom
 - these can be used to develop confidence bounds or hypothesis tests for μ and σ^2 respectively
- the **Student's t distribution** with n degrees of freedom, denoted Student- $t(n)$, has the following properties:
 - it looks like a standard normal as $n \rightarrow \infty$
 - is symmetric about 0
 - has mean $\mathbb{E}_{\text{Stu}_n}[X] = 0$ for $n > 1$
 - * mean undefined for $n = 1$
 - has variance $\mathbb{V}_{\text{Stu}_n}[X] = \frac{n}{n-2}$ for $n > 2$

* variance undefined for $n \leq 2$

- the **chi-squared distribution** with n degrees of freedom, denoted χ_n^2 , has the following properties:
 - its samples are always positive
 - mean and variance $\mathbb{E}_{\chi_n^2}[X] = n$, $\mathbb{V}_{\chi_n^2}[X] = 2n$
 - chi-squared variables add if you also add their degrees of freedom
 - * thus if $X_1 \sim \chi_{n_1}^2$ and $X_2 \sim \chi_{n_2}^2$ then $X_1 + X_2 \sim \chi_{n_1+n_2}^2$

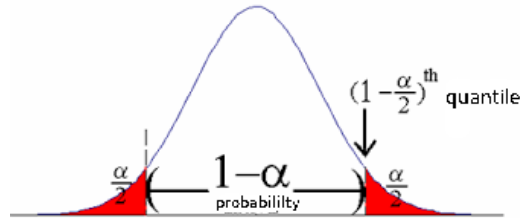
6 Confidence Intervals

- let X have the CDF $P(X)$, and $Q(p) = P^{-1}(p)$ is the corresponding quantile function; then the $(1 - \alpha)$ **two-sided confidence interval** for X is given by

$$[Q(\alpha/2), Q(1 - \alpha/2)]$$

- consider the case for $Z \sim N(0, 1)$:
 - let $Z_{1-\alpha/2}$ denotes the upper $\alpha/2$ quantile for $N(0, 1)$
 - we are $1 - \alpha$ confident $Z \sim N(0, 1)$ falls inside $(-Z_{1-\alpha/2}, Z_{1-\alpha/2})$
 - $[-Z_{1-\alpha/2}, Z_{1-\alpha/2}]$ is called a (two-sided) confidence interval for $N(0, 1)$

this is depicted in the unshaded part of the curve:



- let X have the CDF $P(X)$, and $Q(p) = P^{-1}(p)$ is the corresponding quantile function; then the $(1 - \alpha)$ **one-sided upper confidence interval** for X is given by

$$[Q(\alpha), -\infty]$$

- assume dataset of count n with mean $\bar{X}$ and sample variance S^2 :

assumptions	parameter	interval
Gaussian, σ^2 known	μ	$\bar{X} \pm Z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$
Gaussian, σ^2 unknown	μ	$\bar{X} \pm t_{\alpha/2, n-1} \frac{S}{\sqrt{n}}$
Gaussian, μ unknown	σ^2	$\left(\frac{(n-1)S^2}{\chi_{\alpha/2, n-1}^2}, \frac{(n-1)S^2}{\chi_{1-\alpha/2, n-1}^2} \right)$

- assume dataset of count n with mean $\bar{X}$ and sample variance S^2 and a second dataset of count m with mean $\bar{Y}$ and sample variance T^2 :

assumptions	parameter	interval
Gaussian, σ_1^2, σ_2^2 known	$\mu_1 - \mu_2$	$\bar{X} - \bar{Y} \pm Z_{\alpha/2} \sqrt{\sigma_1^2/n + \sigma_2^2/m}$
Gaussian, $\sigma_1^2 = \sigma_2^2$ unknown but equal	$\mu_1 - \mu_2$	$\bar{X} - \bar{Y} \pm t_{\alpha/2, n+m-2} \sqrt{\frac{1}{n} + \frac{1}{m}} S_P$ for $S_P^2 = \frac{(n-1)S^2 + (m-1)T^2}{n+m-2}$
Gaussian, $\sigma_1^2 \neq \sigma_2^2$ unknown, using CLT	$\mu_1 - \mu_2$	use 1st case for $\sigma_1^2 = S^2, \sigma_2^2 = T^2$, assuming n, m are large

- for Poisson, assume dataset of count n with mean $\hat{X}$; for Bernoulli, assume dataset of count n with mean $\hat{p}$; also a 2nd dataset of count m with mean $\hat{q}$;

assumptions	parameter	interval
Poisson, λ unknown, using CLT	λ	$\hat{X} \pm Z_{\alpha/2} \sqrt{\hat{X}/n}$
Bernoulli, θ unknown, using CLT	θ	$\hat{p} \pm Z_{\alpha/2} \sqrt{\hat{p}(1-\hat{p})/n}$
Bernoulli, θ_1, θ_2 unknown, using CLT	$\theta_1 - \theta_2$	$\hat{p} - \hat{q} \pm Z_{\alpha/2} \sqrt{\hat{p}(1-\hat{p})/n + \hat{q}(1-\hat{q})/m}$

7 Hypothesis Tests

- given a **test statistic** t with CDF $P(T)$, then the p-value is given by

$$p = \begin{cases} 2 \min(P(t), 1 - P(t)) & \text{if null hypothesis is equality} \\ 1 - P(t) & \text{if null hypothesis is lower bound} \\ P(t) & \text{if null hypothesis is upper bound} \end{cases}$$

- assume dataset of count n with mean $\bar{X}$ and sample variance S^2 :

assumptions	null-hypo.	test statistic
Gaussian, σ^2 known	μ_0	$Z = \frac{\bar{X} - \mu_0}{\sigma/\sqrt{n}}$
Gaussian, σ^2 unknown	μ_0	$t_{n-1} = \frac{\bar{X} - \mu_0}{S/\sqrt{n}}$
Gaussian, μ unknown	σ_0^2	$\chi_{n-1}^2 = \frac{(n-1)S^2}{\sigma_0^2}$

- assume dataset of count n with mean $\bar{X}$ and sample variance S^2 and a second dataset of count m with mean $\bar{Y}$ and sample variance T^2 :

assumptions	null-hypo.	test statistic
Gaussian, σ_1^2, σ_2^2 known	$\Delta\mu_0$	$Z = \frac{\bar{X} - \bar{Y} - \Delta\mu_0}{\sqrt{\sigma_1^2/n + \sigma_2^2/m}}$
Gaussian, $\sigma_1^2 = \sigma_2^2$ unknown but equal	$\Delta\mu_0$	$t_{n+m-2} = \frac{\bar{X} - \bar{Y} - \Delta\mu_0}{S_P \sqrt{\frac{1}{n} + \frac{1}{m}}}$ for $S_P^2 = \frac{(n-1)S^2 + (m-1)T^2}{n+m-2}$
Gaussian, $\sigma_1^2 \neq \sigma_2^2$ unknown, using CLT	$\Delta\mu_0$	use 1st case for $\sigma_1^2 = S^2, \sigma_2^2 = T^2$, assuming n, m are large

- for Poisson, assume dataset of count n with mean $\hat{X}$; for Bernoulli, assume dataset of count n with mean $\hat{p}$; also a 2nd dataset of count m with mean $\hat{q}$; all using the CLT so require large samples (n, m)

assumptions	null-hypo.	test statistic
Poisson, λ unknown, using CLT	λ_0	$Z = \frac{\hat{X} - \lambda_0}{\sqrt{\hat{X}/n}}$
Bernoulli, θ unknown, using CLT	θ_0	$Z = \frac{\hat{p} - \theta_0}{\sqrt{\hat{p}(1-\hat{p})/n}}$
Bernoulli, θ_1, θ_2 unknown, using CLT	$\Delta\theta_0$	$Z = \frac{\hat{p} - \hat{q} - \Delta\theta_0}{\sqrt{\hat{p}(1-\hat{p})/n + \hat{q}(1-\hat{q})/m}}$

8 Linear Regression

- simple least squares model has $\mathbb{E}[y_i | x_i] = \beta_0 + \beta_1 x_i$ and residual sum of squares (RSS) of

$$\text{RSS}(\beta_0, \beta_1) = \sum_{i=1}^n (y_i - \beta_0 - \beta_1 x_i)^2$$

which when minimised gives the solution for β_0 of

$$\hat{\beta}_0 = \frac{\overline{Y} \overline{X^2} - \overline{XY} \overline{X}}{\overline{X^2} - \overline{X}^2}$$

and the solution for β_1 of

$$\hat{\beta}_1 = \frac{\overline{XY} - \hat{\beta}_0 \overline{X}}{\overline{X^2}}$$

- if we use the probability model $y_i \sim N(\beta_0 + \beta_1 x_i, \sigma^2)$ then the log-likelihood becomes

$$L(\mathbf{x}, \mathbf{y} | \beta_0, \beta_1, \sigma^2) = \frac{n}{2} \log(2\pi\sigma^2) + \frac{\text{RSS}(\beta_0, \beta_1)}{2\sigma^2}$$

- minimising this gives the same solution to β_0, β_1 as before and an estimator for σ^2

$$\hat{\sigma}_{ML}^2 = \frac{1}{n} \text{RSS}(\hat{\beta}_0, \hat{\beta}_1)$$

plus an unbiased estimator of σ^2 is given by

$$\hat{\sigma}_u^2 = \frac{1}{n-2} \text{RSS}(\hat{\beta}_0, \hat{\beta}_1)$$

- moreover, the following statistics can be used to develop confidence intervals or hypothesis tests

$$\begin{aligned} \frac{1}{\sigma^2} \text{RSS}(\hat{\beta}_0, \hat{\beta}_1) &\sim \chi_{n-2}^2 \\ \frac{1}{\sqrt{\frac{\text{RSS}}{n(n-2)} \frac{\bar{X}^2}{\bar{X}^2 - \bar{X}^2}}} (\hat{\beta}_0 - \beta_0) &\sim \text{Student-t}(n-2) \\ \frac{1}{\sqrt{\frac{\text{RSS}}{n(n-2)} \frac{1}{\bar{X}^2 - \bar{X}^2}}} (\hat{\beta}_1 - \beta_1) &\sim \text{Student-t}(n-2) \end{aligned}$$

- a measure of quality for the linear regression is the $\mathbf{R}^2$ value computed as

$$\begin{aligned} T^2 &= \sum_{i=1}^n (y_i - \bar{y})^2 \\ R^2 &= 1 - \frac{\text{RSS}}{T^2} \end{aligned}$$

which is in $[0, 1]$, 1 for a perfect zero error fit, 0 for pure noise, and higher for better quality fit

- for **multi-linear regression**, the prediction instead becomes

$$\mathbb{E}[y_i | x_{i,1}, \dots, x_{i,p}] = \beta_0 + \sum_{j=1}^p \beta_j x_{i,j}$$

and residual sum of squares (RSS) of

$$\text{RSS}(\beta_0, \beta_1, \dots, \beta_p) = \sum_{i=1}^n \left(y_i - \beta_0 - \sum_{j=1}^p \beta_j x_{i,j} \right)^2$$

- the **design matrix** given by $\mathbf{X}$ of predictors:

$$\mathbf{X} = (\mathbf{1}, \mathbf{x}'_1, \mathbf{x}'_2, \dots, \mathbf{x}'_p) = \begin{pmatrix} 1 & x_{1,1} & x_{1,2} & \cdots & x_{1,p} \\ 1 & x_{2,1} & x_{2,2} & \cdots & x_{2,p} \\ \vdots & \vdots & \vdots & & \vdots \\ 1 & x_{n,1} & x_{n,2} & \cdots & x_{n,p} \end{pmatrix},$$

has corresponding parameters $\boldsymbol{\beta}^T = (\beta_0, \beta_1, \dots, \beta_p)$ yielding the prediction

$$\mathbb{E}[y_i | \mathbf{x}_i] = \mathbf{X}\boldsymbol{\beta}$$

- minimising $\text{RSS}(\boldsymbol{\beta})$ has solution

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}'\mathbf{X})^{-1}(\mathbf{X}'\mathbf{Y})$$

- if we use the probability model $y_i \sim N(\mathbf{x}_i\boldsymbol{\beta}, \sigma^2)$ then the log-likelihood becomes

$$L(\mathbf{X}, \mathbf{y} | \boldsymbol{\beta}, \sigma^2) = \frac{n}{2} \log(2\pi\sigma^2) + \frac{\text{RSS}(\boldsymbol{\beta})}{2\sigma^2}$$

- minimising this gives the same solution to $\boldsymbol{\beta}$ as before and an estimator for σ^2

$$\hat{\sigma}_{ML}^2 = \frac{1}{n} \text{RSS}(\hat{\boldsymbol{\beta}})$$

plus an unbiased estimator of σ^2 is given by

$$\hat{\sigma}_u^2 = \frac{1}{n-p-1} \text{RSS}(\hat{\boldsymbol{\beta}})$$

9 Classification

- probability prediction formula for **naïve Bayes classifier** is

$$P(y | x_1, \dots, x_p) = \frac{P(y) \prod_{j=1}^p P(x_j | y)}{P(x_1, \dots, x_p)}$$

where the denominator is a constant so can be found by normalising the numerator.

- point estimation is done by estimating the probabilities $P(Y = y)$ and $P(X_j = x_j | Y = y)$ for all entries of the tables

- probability prediction formula for **logistic regression classifier** is expressed using the **logistic function**

$$p(Y_i = 1 \mid x_{i,1}, \dots, x_{i,p}) = \frac{1}{1 + \exp(\eta_i)}$$

where

$$\eta_i = \beta_0 + \sum_{j=1}^p \beta_j x_{i,j}$$

so that the **log-odds** given by

$$\log \frac{p(Y_i = 1 \mid x_{i,1}, \dots, x_{i,p})}{p(Y_i = 0 \mid x_{i,1}, \dots, x_{i,p})} = \eta_i$$

- the parameters $(\beta_0, \beta_1, \dots, \beta_p)$ are fit using optimising routines on the log likelihood

10 Table for Standard Normal

Table by LawrenceSeminarioRomero [CC BY-SA 4.0], from Wikimedia Commons. $F(Z)$ gives the CDF for the standard normal Gaussian.

Table of Standard Normal Probabilities							
Z	F(Z)	Z	F(Z)	Z	F(Z)	Z	F(Z)
-3.00	0.0013	-1.48	0.0694	0.04	0.5160	1.56	0.9406
-2.96	0.0015	-1.44	0.0749	0.08	0.5319	1.60	0.9452
-2.92	0.0018	-1.40	0.0808	0.12	0.5478	1.64	0.9495
-2.88	0.0020	-1.36	0.0869	0.16	0.5636	1.68	0.9535
-2.84	0.0023	-1.32	0.0934	0.20	0.5793	1.72	0.9573
-2.80	0.0026	-1.28	0.1003	0.24	0.5948	1.76	0.9608
-2.76	0.0029	-1.24	0.1075	0.28	0.6103	1.80	0.9641
-2.72	0.0033	-1.20	0.1151	0.32	0.6255	1.84	0.9671
-2.68	0.0037	-1.16	0.1230	0.36	0.6406	1.88	0.9699
-2.64	0.0041	-1.12	0.1314	0.40	0.6554	1.92	0.9726
-2.60	0.0047	-1.08	0.1401	0.44	0.6700	1.96	0.9750
-2.56	0.0052	-1.04	0.1492	0.48	0.6844	2.00	0.9772
-2.52	0.0059	-1.00	0.1587	0.52	0.6985	2.04	0.9793
-2.48	0.0066	-0.96	0.1685	0.56	0.7123	2.08	0.9812
-2.44	0.0073	-0.92	0.1788	0.60	0.7257	2.12	0.9830
-2.40	0.0082	-0.88	0.1894	0.64	0.7389	2.16	0.9846
-2.36	0.0091	-0.84	0.2005	0.68	0.7517	2.20	0.9861
-2.32	0.0102	-0.80	0.2119	0.72	0.7642	2.24	0.9875
-2.28	0.0113	-0.76	0.2236	0.76	0.7764	2.28	0.9887
-2.24	0.0125	-0.72	0.2358	0.80	0.7881	2.32	0.9898
-2.20	0.0139	-0.68	0.2483	0.84	0.7995	2.36	0.9909
-2.16	0.0154	-0.64	0.2611	0.88	0.8106	2.40	0.9918
-2.12	0.0170	-0.60	0.2743	0.92	0.8212	2.44	0.9927
-2.08	0.0188	-0.56	0.2877	0.96	0.8315	2.48	0.9934
-2.04	0.0207	-0.52	0.3015	1.00	0.8413	2.52	0.9941
-2.00	0.0228	-0.48	0.3156	1.04	0.8508	2.56	0.9948
-1.96	0.0250	-0.44	0.3300	1.08	0.8599	2.60	0.9953
-1.92	0.0274	-0.40	0.3446	1.12	0.8686	2.64	0.9959
-1.88	0.0301	-0.36	0.3594	1.16	0.8770	2.68	0.9963
-1.84	0.0329	-0.32	0.3745	1.20	0.8849	2.72	0.9967
-1.80	0.0359	-0.28	0.3897	1.24	0.8925	2.76	0.9971
-1.76	0.0392	-0.24	0.4052	1.28	0.8997	2.80	0.9974
-1.72	0.0427	-0.20	0.4207	1.32	0.9066	2.84	0.9977
-1.68	0.0465	-0.16	0.4364	1.36	0.9131	2.88	0.9980
-1.64	0.0505	-0.12	0.4522	1.40	0.9192	2.92	0.9982
-1.60	0.0548	-0.08	0.4681	1.44	0.9251	2.96	0.9985
-1.56	0.0594	-0.04	0.4840	1.48	0.9306	3.00	0.9987
-1.52	0.0643	0.00	0.5000	1.52	0.9357		

11 Table for Student t

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$\begin{matrix} p \\ \text{df} \end{matrix}$	0,25	0,2	0,15	0,1	0,05	0,025	0,01	0,005	0,0005
1	1,000	1,376	1,963	3,078	6,314	12,706	31,821	63,656	636,578
2	0,816	1,061	1,386	1,886	2,920	4,303	6,965	9,925	31,600
3	0,765	0,978	1,250	1,638	2,353	3,182	4,541	5,841	12,924
4	0,741	0,941	1,190	1,533	2,132	2,776	3,747	4,604	8,610
5	0,727	0,920	1,156	1,476	2,015	2,571	3,365	4,032	6,869
6	0,718	0,906	1,134	1,440	1,943	2,447	3,143	3,707	5,959
7	0,711	0,896	1,119	1,415	1,895	2,365	2,998	3,499	5,408
8	0,706	0,889	1,108	1,397	1,860	2,306	2,896	3,355	5,041
9	0,703	0,883	1,100	1,383	1,833	2,262	2,821	3,250	4,781
10	0,700	0,879	1,093	1,372	1,812	2,228	2,764	3,169	4,587
11	0,697	0,876	1,088	1,363	1,796	2,201	2,718	3,106	4,437
12	0,695	0,873	1,083	1,356	1,782	2,179	2,681	3,055	4,318
13	0,694	0,870	1,079	1,350	1,771	2,160	2,650	3,012	4,221
14	0,692	0,868	1,076	1,345	1,761	2,145	2,624	2,977	4,140
15	0,691	0,866	1,074	1,341	1,753	2,131	2,602	2,947	4,073
16	0,690	0,865	1,071	1,337	1,746	2,120	2,583	2,921	4,015
17	0,689	0,863	1,069	1,333	1,740	2,110	2,567	2,898	3,965
18	0,688	0,862	1,067	1,330	1,734	2,101	2,552	2,878	3,922
19	0,688	0,861	1,066	1,328	1,729	2,093	2,539	2,861	3,883
20	0,687	0,860	1,064	1,325	1,725	2,086	2,528	2,845	3,850
21	0,686	0,859	1,063	1,323	1,721	2,080	2,518	2,831	3,819
22	0,686	0,858	1,061	1,321	1,717	2,074	2,508	2,819	3,792
23	0,685	0,858	1,060	1,319	1,714	2,069	2,500	2,807	3,768
24	0,685	0,857	1,059	1,318	1,711	2,064	2,492	2,797	3,745
25	0,684	0,856	1,058	1,316	1,708	2,060	2,485	2,787	3,725
26	0,684	0,856	1,058	1,315	1,706	2,056	2,479	2,779	3,707
27	0,684	0,855	1,057	1,314	1,703	2,052	2,473	2,771	3,689
28	0,683	0,855	1,056	1,313	1,701	2,048	2,467	2,763	3,674
29	0,683	0,854	1,055	1,311	1,699	2,045	2,462	2,756	3,660
30	0,683	0,854	1,055	1,310	1,697	2,042	2,457	2,750	3,646
40	0,681	0,851	1,050	1,303	1,684	2,021	2,423	2,704	3,551
60	0,679	0,848	1,045	1,296	1,671	2,000	2,390	2,660	3,460
120	0,677	0,845	1,041	1,289	1,658	1,980	2,358	2,617	3,373
∞	0,674	0,842	1,036	1,282	1,645	1,960	2,326	2,576	3,290

12 Table for Chi-squared

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Critical Values of the χ^2 Distribution										
df \ α	0.995	0.975	0.9	0.5	0.1	0.05	0.025	0.01	0.005	df
1	.000	.000	0.016	0.455	2.706	3.841	5.024	6.635	7.879	1
2	0.010	0.051	0.211	1.386	4.605	5.991	7.378	9.210	10.597	2
3	0.072	0.216	0.584	2.366	6.251	7.815	9.348	11.345	12.838	3
4	0.207	0.484	1.064	3.357	7.779	9.488	11.143	13.277	14.860	4
5	0.412	0.831	1.610	4.351	9.236	11.070	12.832	15.086	16.750	5
6	0.676	1.237	2.204	5.348	10.645	12.592	14.449	16.812	18.548	6
7	0.989	1.690	2.833	6.346	12.017	14.067	16.013	18.475	20.278	7
8	1.344	2.180	3.490	7.344	13.362	15.507	17.535	20.090	21.955	8
9	1.735	2.700	4.168	8.343	14.684	16.919	19.023	21.666	23.589	9
10	2.156	3.247	4.865	9.342	15.987	18.307	20.483	23.209	25.188	10
11	2.603	3.816	5.578	10.341	17.275	19.675	21.920	24.725	26.757	11
12	3.074	4.404	6.304	11.340	18.549	21.026	23.337	26.217	28.300	12
13	3.565	5.009	7.042	12.340	19.812	22.362	24.736	27.688	29.819	13
14	4.075	5.629	7.790	13.339	21.064	23.685	26.119	29.141	31.319	14
15	4.601	6.262	8.547	14.339	22.307	24.996	27.488	30.578	32.801	15